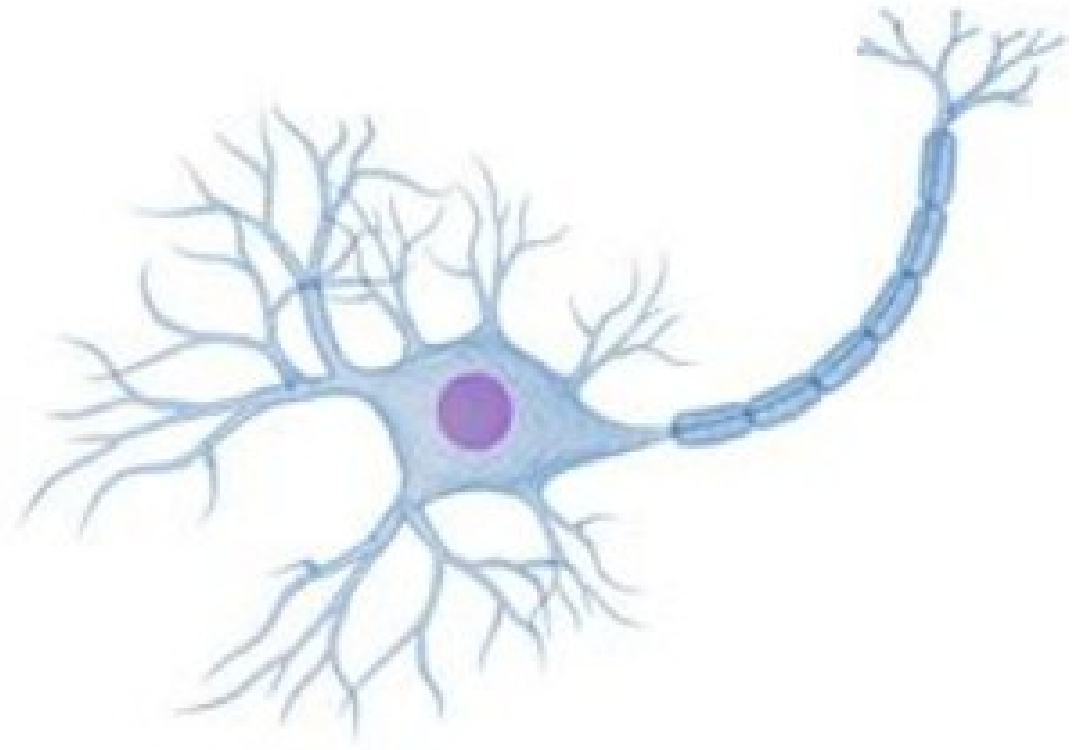
Neurons



Neurons Communicate Quickly

Nervous System Overview

- Cells, tissues and organs need to work in an integrated, coordinated way to maintain homeostasis.
- Nervous System (NS) allows this communication in a fast, responsive way.
- Impulses passed along specialised cells called *neurons*.

Sensory NS – takes information *from the body to the brain* about:

- External conditions: temperature, vision, hearing, touch, taste etc.
- Internal conditions: blood pressure, blood gas and pH, stretch of intestines etc.

Central NS – brain and spinal cord

- Spinal cord – takes information to and from brain, coordinates spinal reflexes
- Brain – processes information and sends out response.

Motor NS - *takes information from brain to body* about response

- Synapses are at muscles and glands
- Stimulates movement, or glandular secretions
- Has both voluntary and involuntary sections

Example: A velociraptor escapes from Jurassic Park and enters the classroom:

Sensory NS: your eyes see the velociraptor and send a signal to the brain.

Central NS: your brain processes the information.

- Consciously notices velociraptor, makes plan to run away, sends signal down spinal cord and out to leg muscles.
- Conscious brain sends signal to hypothalamus to trigger "fight or flight" response.

Motor NS:

- Carries conscious (somatic, voluntary) signal to skeletal system to move leg muscles to run away (or to use your friend as a shield)

- Carries unconscious (autonomic, involuntary) signal to adrenal cortex to release adrenalin, which increases blood pressure, dilates respiratory passages … and so on.



Nerve Cells (neurons)

- Are the structural and functional unit of the nervous system
- Vary in size and shape

COMMON FEATURES

Cell body:

- Contains nucleus and organelles

Dendrites:

- Short extensions of cell body
- Highly branched
- Receive stimulus and carry impulses to cell body

Axon (nerve fibre)

- Longer extension
- Carries nerve impulses away from cell body
- Usually covered with myelin sheath (*myelinated)*
- *Unmyelinated* = no myelin

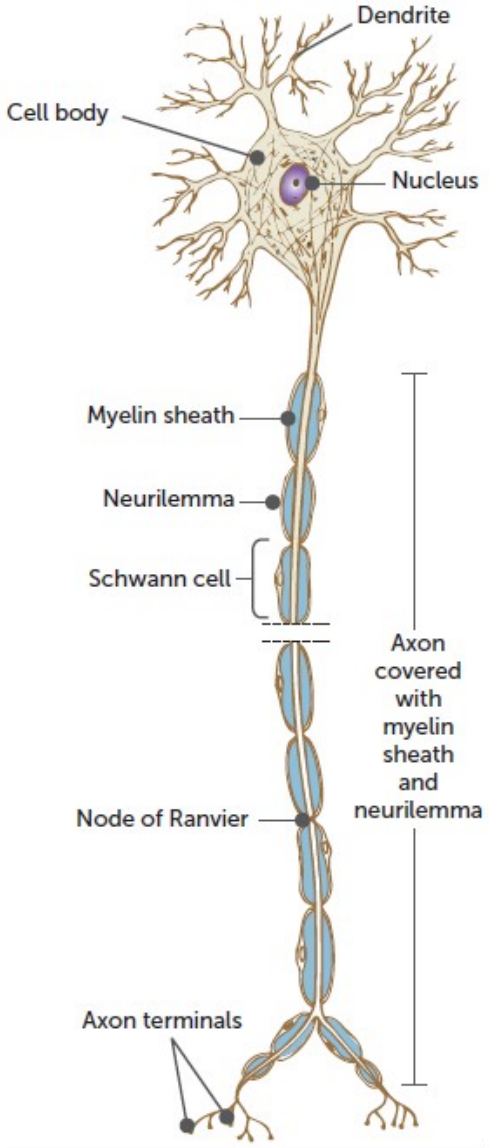


FIGURE 3.2 Structure of a typical neuron with a myelinated axon

Learning Aim: Describe the structure of a neuron, with location and function of common features

The Myelin Sheath and Neurilemma

- Most neurons are *myelinated* – they have a myelin sheath:
  - Formed by specialised cells:
    - *Schwann Cells* – peripheral NS
    - *Oligodendrocytes* – central NS – myelin forms "white matter".
  - Gaps along axon: *nodes of Ranvier*
  - 3 functions of myelin sheath:
    - Acts as an insulator for impulses
    - Protects axon from external damage
    - Speeds movement of nerve impulses
  - *Neurilemma*: the outermost layer of Schwann cells – helps repair injured fibres by guiding regrowth.

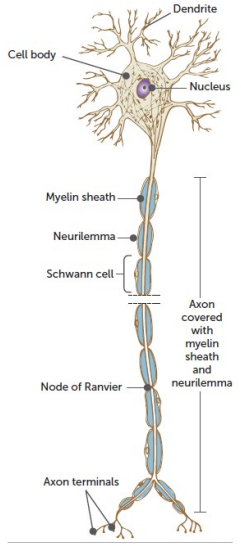


FIGURE 3.2 Structure of a typical neuron with a myelinated axon

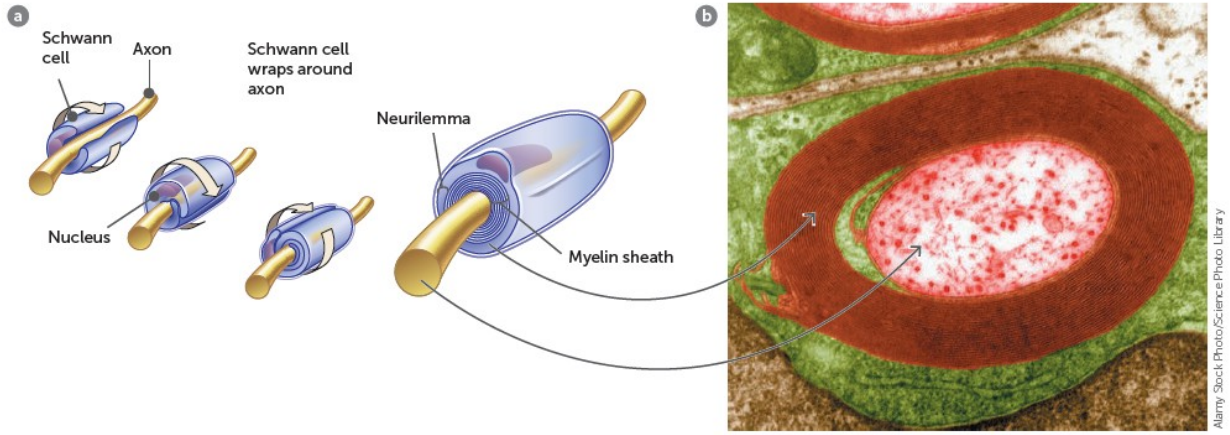
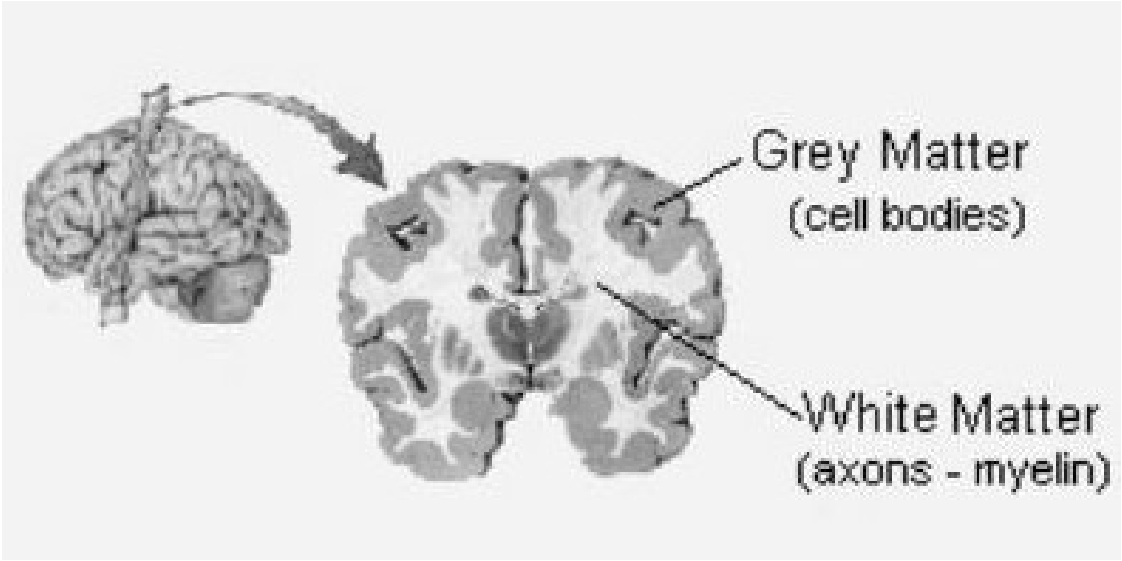


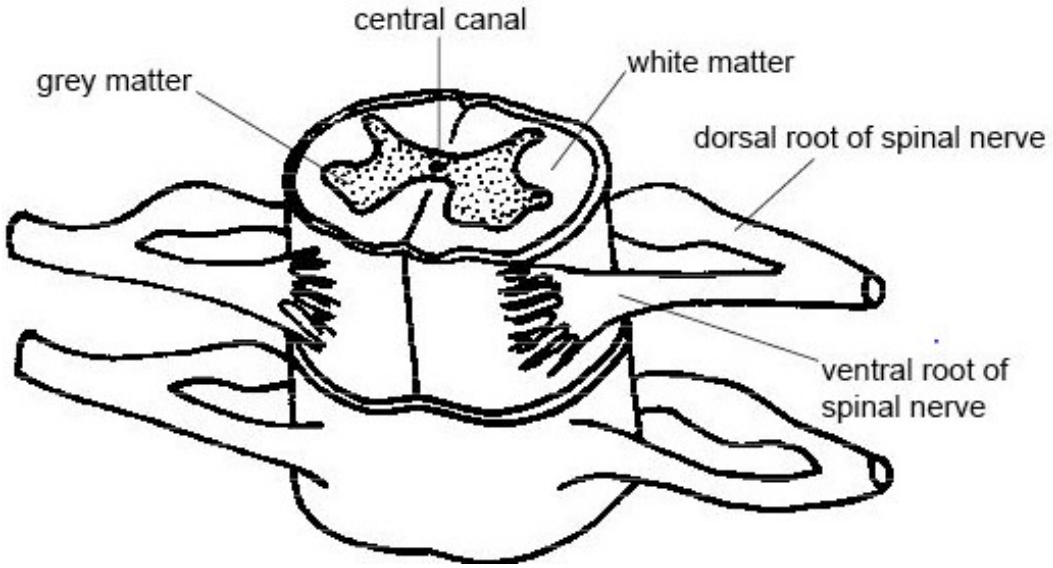
FIGURE 3.3 a Schwann cells form the sheath of a myelinated axon by wrapping around the axon. **b** Coloured transmission electron micrograph of a transverse section of a myelinated axon. The myelin sheath (brown) is formed when Schwann cells wrap around the axon and deposit layers of myelin between each coil. The outermost layer (green) is the cytoplasm of the Schwann cell – the neurilemma

Learning Aim: Describe the structure and functions of the myelin sheath

When CNS is cut open:

- Grey matter:
  - Cell bodies
  - Unmyelinated fibres
- White matter:
  - Myelinated fibres





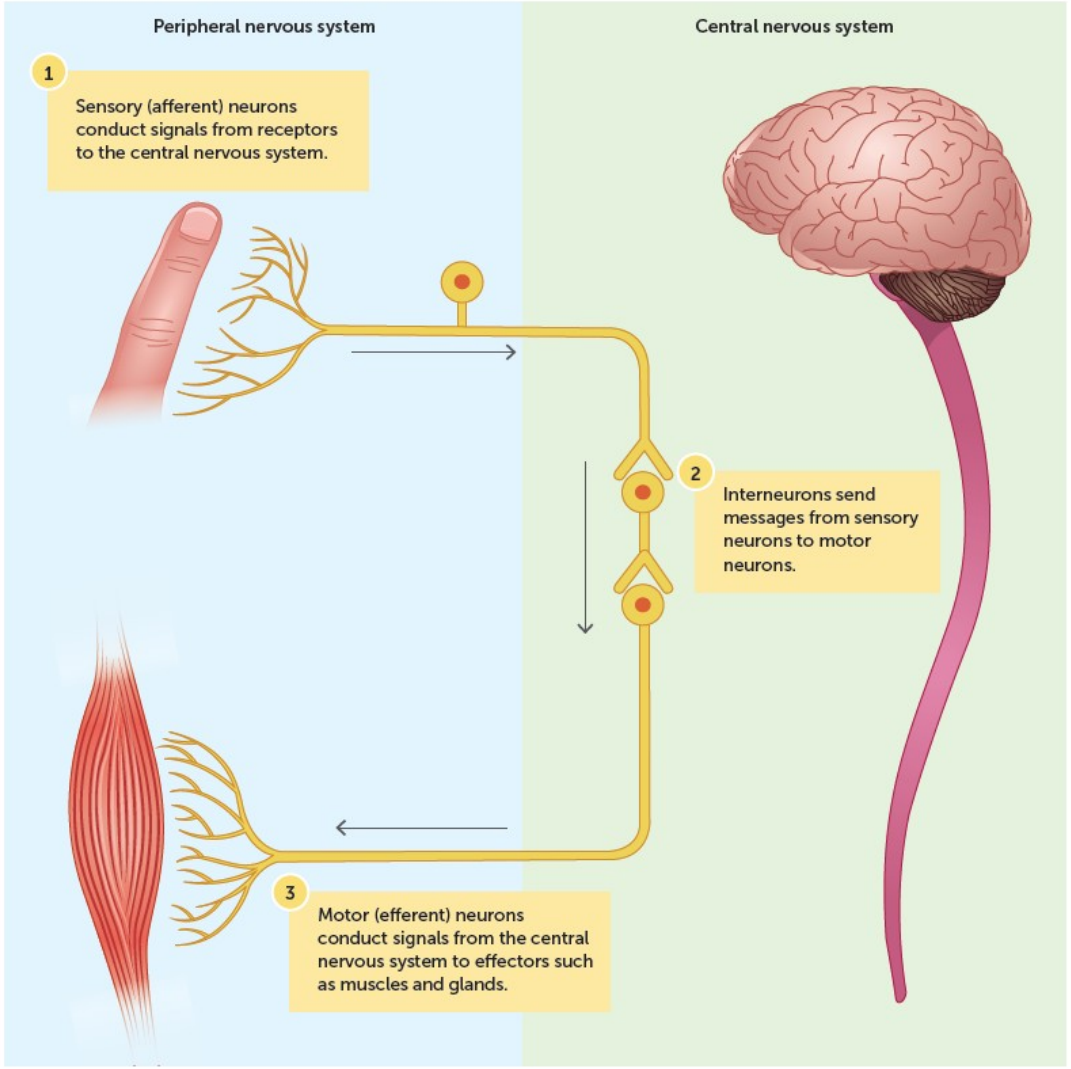
Types of neurons - Function

Neurons can be classified by function:

- **Sensory neurons**
  - Also called afferent neurons, or receptor neurons
  - Carry messages from receptors to CNS

- **Interneurons**
  - Also called association neurons, or relay neurons
  - Link between sensory and motor neurons
  - Generally in Central Nervous System (CNS)

- **Motor neurons**
  - Also called effector neurons
  - Carry messages from CNS to muscles and glands



- Can be *somatic* – under voluntary control, eg making skeletal muscle move
- Can be *autonomic* – involuntary, eg controlling muscle lining blood vessels and glands

Learning Aim: Describe the three functional types of neuron

Types of neurons - Structure

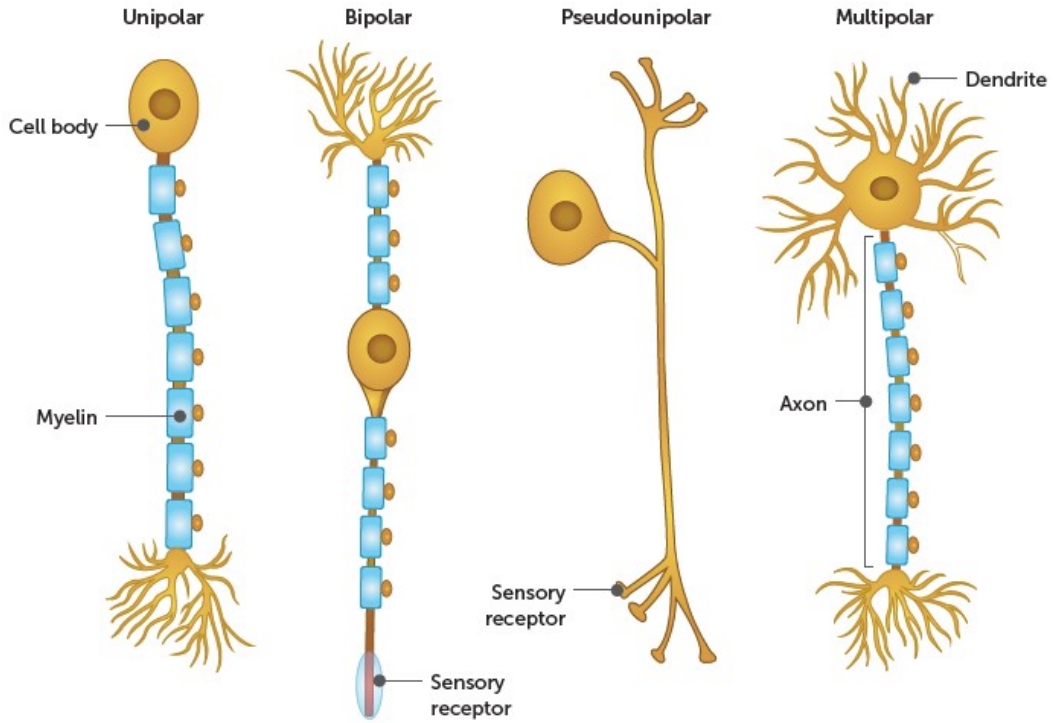
Neurons can be classified by structure:

- Multipolar
  - Most common
  - One axon, multiple dendrites
  - Axon may have multiple terminals
  - Interneurons and motor neurons are this type

- Bipolar neurons
  - One axon, one dendrite, branching at ends
  - Generally part of special sense organs
  - Eg. in eye, ear, nose: take impulses from receptor to other neurons

- Pseudounipolar neurons
  - Cell body to side
  - Most sensory neurons are this type
  - Sometimes (not really correctly) called unipolar

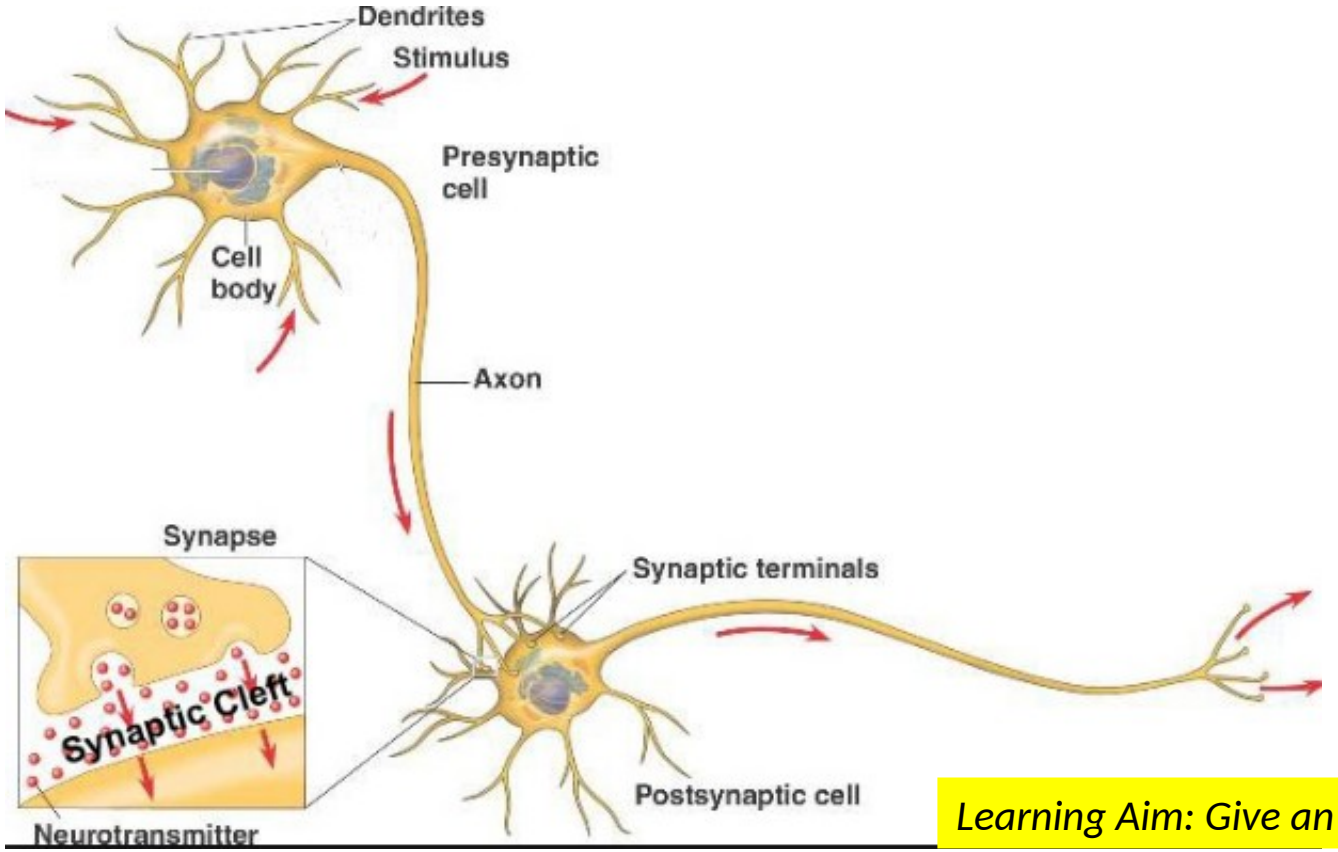
- Unipolar neurons
  - No dendrites
  - Uncommon in vertebrates such as humans

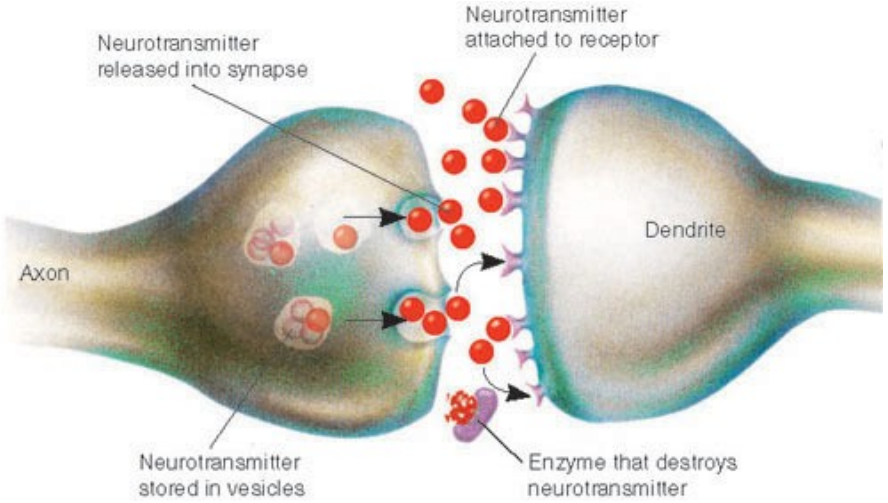


Learning Aim: Describe the main structural types of neuron and describe where each occur

Synapses

- Point where nerve impulses are passed from one neuron to another.
- No join: small gap for chemical messages *neurotransmitters* to cross.
- Also occur between nerve and muscle: *neuromuscular junction*





Learning Aim: Give an overview of the structure and function of the synapse

Nerves

- Are bundles of neurons (nerve fibres) extending out from the central nervous system.
- Held together by connective tissue sheath
- Three broad types:

SENSORY NERVES

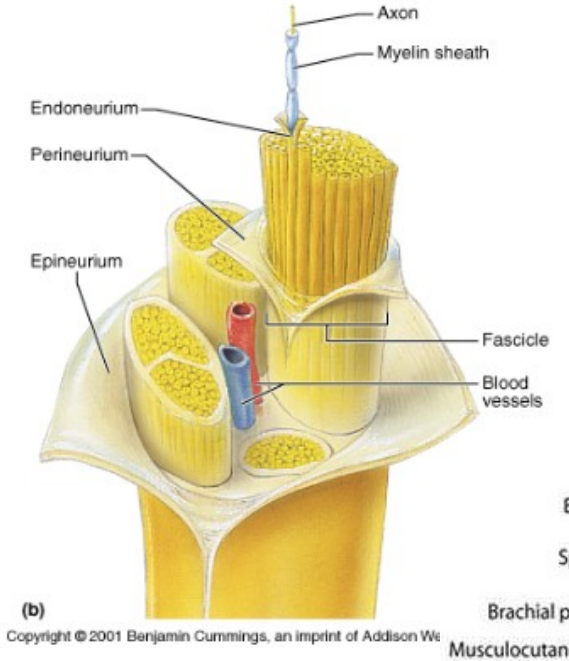
- Neurons are all sensory neurons
- Take sensory information from the body to the CNS (Central Nervous System)

MOTOR NERVES

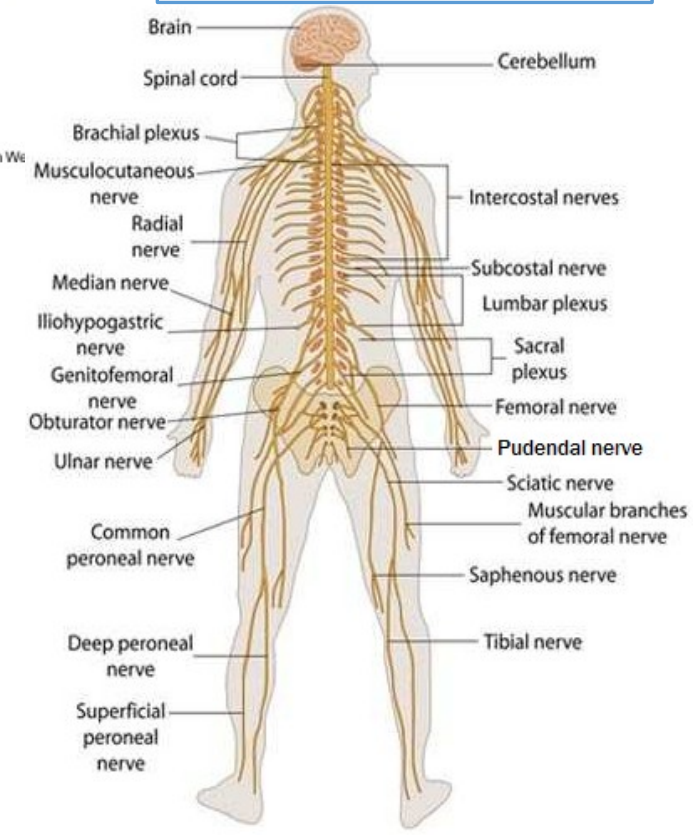
- Neurons are all motor neurons
- Take motor information (to tell body to move) from the CNS to the body

MIXED NERVES

- Contain some bundles of sensory neurons and some bundles of motor neurons
- Take information both to and from the CNS.



Sensory and motor neurons are often bundled together into a single nerve that takes sensory information to the brain, and carries signals from the brain to move muscle/glands.



Learning Aim: Understand the structure of nerves and describe the three main types

Transmission of Nerve Impulses (overview)

Impulses are transmitted along nerves using a wave of electro-chemical depolarisation called an *action potential*

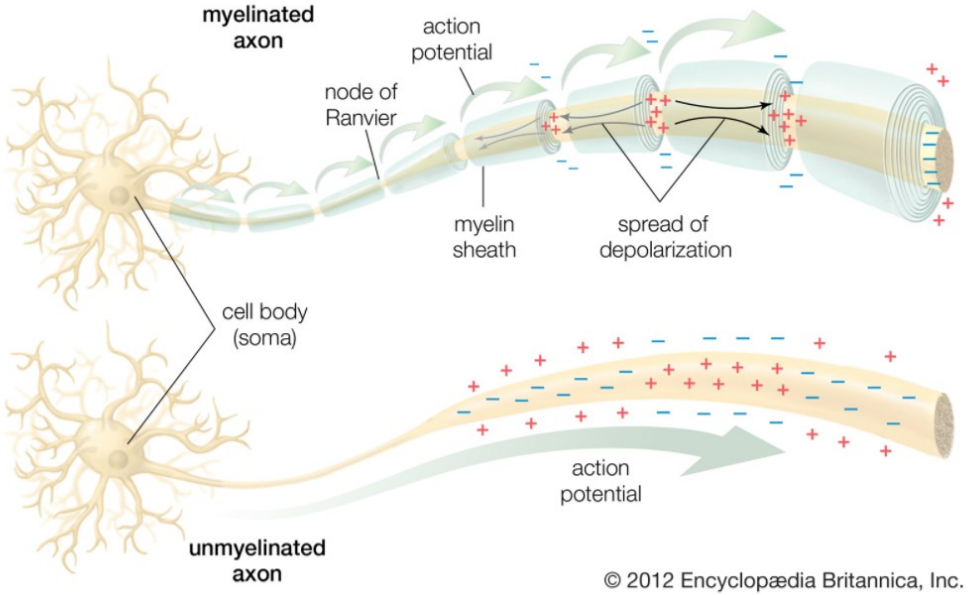
1: Dendrites are stimulated, so they trigger an action potential.

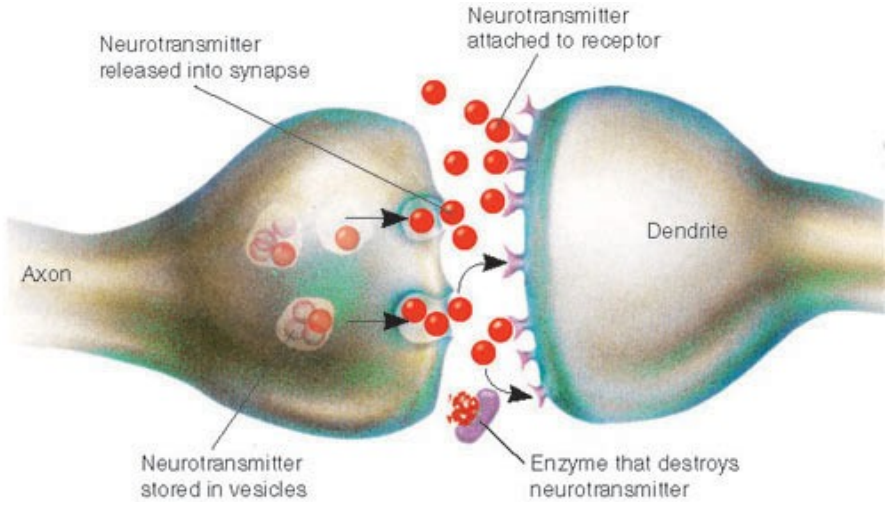
2: Action potential spreads down the neuron:

dendrites → cell body → axon

3: Action potential reaches axon terminal and triggers release of *neurotransmitters* that cross the synapse.

4: Neurotransmitters bind to the dendrites on the other side of the synapse, stimulating an action potential in the next neuron.





Learning Aim: Give an overview of how nerve impulses are transmitted